

The Fundamental Theorem of Calculus

MATH 145 – Applied Calculus

Torin Quinlivan

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Velocity and Distance

Suppose $v(t)$ is the velocity of an object at time t and consider the distance traveled over the time interval Δt .

$$\begin{aligned}\int_a^b v(t) dt &= \lim_{n \rightarrow \infty} \sum_{i=1}^n v(t_i) \Delta t \\ &= \text{distance traveled (displacement)} \\ &= \text{area under the graph of } v(t) \\ &= f(b) - f(a)\end{aligned}$$

The Fundamental Theorem of Calculus

The Fundamental Theorem of Calculus

Suppose $f(t)$ is a continuous function on the interval $[a, b]$ and that $F(t)$ is an antiderivative of $f(t)$. Then

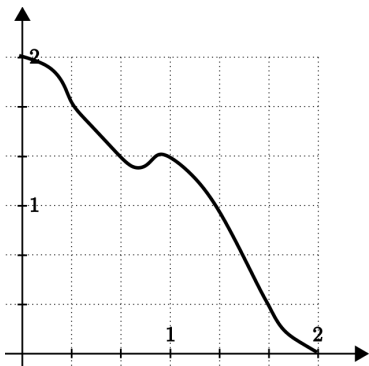
$$\int_a^b f(t) dt = F(b) - F(a)$$

Notes:

- This makes evaluating definite integrals easy, *provided you can find an antiderivative.*
- Alternatively, $\int_a^b f'(t) dt = f(b) - f(a)$.
- Also, $F(b) = F(a) + \int_a^b f(t) dt$.

Example: The Fundamental Theorem of Calculus

Example: At 10 a.m. A cistern at a rate of $R(t)$ gallons/hour, where $R(t)$ has the graph given below.



Estimate the amount of water in the tank at noon.

Example: The Fundamental Theorem of Calculus

Examples:

■ Find $\int_0^2 3x + 1 \, dx$.

■ Find $\int_1^3 (x - 1)^2 \, dx$.

Example: The Fundamental Theorem of Calculus

Examples:

■ Find $\int_0^{\frac{\pi}{2}} \sin(x) dx$.

■ Find $\int_0^3 e^x dx$.

■ Find $\int_0^{\pi} \cos(x) + \frac{1}{\sqrt{1-x^2}} dx$.

Some Theorems for Definite Integrals

$$\blacksquare \int_a^b c \cdot f(x) + d \cdot g(x) dx = c \int_a^b f(x) dx + d \int_a^b g(x) dx.$$

$$\blacksquare \text{ If } a < c < b, \int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx.$$

Note: There is no product or quotient rule for definite integrals.

The Fundamental Theorem of Calculus

Every continuous function $f(x)$ has an antiderivative.

The Fundamental Theorem of Calculus

If $f(x)$ is a continuous function on an interval and a is any point, then the function:

$$F(x) = \int_a^x f(t) dt,$$

is an antiderivative of $f(x)$. In other words:

$$\frac{d}{dx} \left[\int_a^x f(t) dt \right] = f(x).$$